

## PERSONAL INFORMATION



## Mihai-Cristian Farcaș

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Date of birth 5 April 2004 | Nationality Romanian

## WORK EXPERIENCE

2026 – Present

## Software Engineer (Freelance), BluePlan

Built and shipped Atlas ([ai.blueplan.ro](https://ai.blueplan.ro)) end to end: retrieval-augmented chat over technical documents with hybrid dense, sparse and image retrieval fused with RRF, reranking and verbatim citations pinned to the source PDF. Multi-tenant, multilingual and in daily production use.

Technologies Python, FastAPI, Qdrant, React

## 2026 Full-Stack Developer, Baseline Studios

Delivered full-stack features across multiple client applications ([stratosphere.vip](https://stratosphere.vip), [pursuitofx.org](https://pursuitofx.org), [kabu.xyz](https://kabu.xyz), [esoftplus-landing.vercel.app](https://esoftplus-landing.vercel.app)) using TypeScript and React.

Technologies TypeScript, React

## 2025 Software Developer, GeoDraw

Developed an AI-powered canvas for sketching and wireframing apps with React, TLDraw and LLM APIs, turning natural language into accurate real-time design output on the canvas.

Technologies React, TLDraw, LLM APIs

April – June 2025

## Software Development Intern, Tvarita

Rebuilt and redesigned a server monitoring site for an optician clinic chain using React and Tailwind CSS, improving the user experience and significantly reducing load times.

Technologies React, Tailwind CSS

## EDUCATION AND TRAINING

2026 – Present

## MSc in Cloud, Network Infrastructure and High-Performance Computation

Babeș-Bolyai University, Faculty of Mathematics and Computer Science, Cluj-Napoca, Romania

Oct 2023 – June 2026

## BSc in Computer Science (English)

Babeș-Bolyai University, Faculty of Mathematics and Computer Science, Cluj-Napoca, Romania

Thesis: Hybrid post-quantum key exchange for WireGuard, the first FIPS 203-compliant hybrid WireGuard implementation on a production Rust codebase.

## PERSONAL PROJECTS

2026

## pq-boringtun

Hybrid post-quantum key exchange for WireGuard: ML-KEM-768 alongside X25519 in the Noise IK handshake, built in Rust on Cloudflare's BoringTun and benchmarked on Apple M4 and Raspberry Pi 5.

[github.com/mihaicristianfarcas/pq-boringtun](https://github.com/mihaicristianfarcas/pq-boringtun)

**Technologies** Rust, ML-KEM-768, X25519, Noise Protocol, WireGuard

## 2026 Supernova

Live-wallpaper engine for macOS in Swift 6: renders looping video, Shadertoy-style GLSL shaders transpiled to Metal and sandboxed web bundles behind desktop icons, with audio, cursor and motion reactivity, per-display assignment and smart playlists.  
supernova-macos.app

**Technologies** Swift 6, Metal, Core Audio, GLSL

## 2026 mihaicristian.dev

Bilingual (EN/RO) personal portfolio site built with Astro: project showcases, long-form writing and a performance-focused SPA-style browsing experience.  
mihaicristian.dev

**Technologies** Astro, TypeScript

## 2025 eBPF Network Analysis System

Network traffic classification for QUIC protocols, inspecting encrypted flows at the kernel boundary using eBPF with C, Rust and Python.

github.com/mihaicristianfarcas/ebpf-network-analysis-system

**Technologies** C, Rust, Python, eBPF

## 2025 nvim-search-and-replace

Project-wide search and replace plugin for Neovim written in Lua: async streaming ripgrep search, live before/after preview, selective replacement and its own undo/redo stack.  
github.com/mihaicristianfarcas/nvim-search-and-replace

**Technologies** Lua, Neovim, ripgrep

## PERSONAL SKILLS

**Mother tongue** Romanian

**Other languages**

|                                                                         | UNDERSTANDING |         | SPEAKING           |                   | WRITING |
|-------------------------------------------------------------------------|---------------|---------|--------------------|-------------------|---------|
|                                                                         | Listening     | Reading | Spoken interaction | Spoken production |         |
| English                                                                 | C2            | C2      | C2                 | C2                | C2      |
| C2 Certificate of Proficiency in English (Cambridge English Assessment) |               |         |                    |                   |         |

Levels: A1 and A2: Basic user – B1 and B2: Independent user – C1 and C2: Proficient user  
[Common European Framework of Reference for Languages](https://www.cambridgeenglish.org/exams/certificate-of-proficiency-in-english/)

**Digital skills**

- Programming languages: Rust, Go, Python, Swift, C/C++, Java, TypeScript/JavaScript, Lua, Shell
- Web development: React, Next.js, Astro, Tailwind CSS, TLDraw, Vercel, Cloudflare Workers
- Backend, databases and ML: FastAPI, PostgreSQL, Qdrant, FAISS, LangChain, PyTorch, LLM APIs
- Systems, networking and security: eBPF, WireGuard and the Noise protocol, ML-KEM post-quantum key exchange, Metal, Core Audio, GLSL shaders
- Tools: Git, Neovim, ripgrep, Linux, macOS